APM1110: Probability and Probability Distributions

Formative Assessment 8 — Solutions

Problem 1 (6 points)

The urn has 10 red, 8 black and 7 green balls, so 25 balls in total. A color is chosen with probability 1/3 each, then 4 balls are drawn without replacement.

(a) P(X = 0)

Condition on the chosen color. X = 0 means none of the 4 balls has the chosen color, so all 4 come from the other balls.

Red chosen: C(15,4)/C(25,4) = 1365/12650

Black chosen: C(17,4)/C(25,4) = 2380/12650

Green chosen: C(18,4)/C(25,4) = 3060/12650

By the law of total probability:

$P(X = 0) = (1/3)\cdot(1365 + 2380 + 3060)/12650 = (1/3)(6805/12650) = 1361/7590$

Answer: P(X = 0) ≈ 0.1793

(b) $P(X_i = 1)$

The i-th ball drawn from an urn without replacement has the same distribution as the first ball drawn, so each draw is red with probability 10/25, black with 8/25, green with 7/25. Then

$P(X_i = 1) = (1/3)(10/25) + (1/3)(8/25) + (1/3)(7/25) = (1/3)(25/25) = 1/3$

Answer: $P(X_i = 1) = 1/3 \approx 0.3333$ for i = 1, 2, 3, 4

(c) E[X]

The number of balls of the chosen color is $X = X_1 + X_2 + X_3 + X_4$. By linearity of expectation, using part (b):

$E[X] = \Sigma\, E[X_i] = \Sigma\, P(X_i = 1) = 4\cdot(1/3) = 4/3$

Answer: E[X] = 4/3 ≈ 1.3333

Problem 2 (8 points)

$P(X = n) = 4 / [n(n+1)(n+2)]$, $n \geq 1$.

(a) Show that Σ P(X = n) = 1

Since the probabilities are positive, we only need the sum. Using the hint:

$1/[n(n+1)(n+2)] = (1/2)\cdot[1/(n(n+1)) - 1/((n+1)(n+2))]$

Check: $1/(n(n+1)) - 1/((n+1)(n+2)) = [(n+2) - n]/[n(n+1)(n+2)] = 2/[n(n+1)(n+2)]$. Hence

$\Sigma_{n=1}^{\infty} P(X = n) = 4\cdot(1/2)\, \Sigma\, [1/(n(n+1)) - 1/((n+1)(n+2))] = 2\, \Sigma\, [a_n - a_{n+1}]$

with $a_n = 1/(n(n+1))$. The series telescopes. The partial sum up to N is $a_1 - a_{N+1}$, and $a_{N+1} \to 0$, so

$\Sigma\ P(X = n) = 2 \cdot a_1 = 2 \cdot (1/2) = 1$

Answer: The probabilities sum to 1, so this is a valid pmf. (QED)

(b) Show that E[X] = 2

$E[X] = \Sigma\ n \cdot 4/[n(n+1)(n+2)] = 4\ \Sigma\ 1/[(n+1)(n+2)]$

Using $1/[(n+1)(n+2)] = 1/(n+1) - 1/(n+2)$ (the hint with k = 1), the sum telescopes:

$E[X] = 4\ \Sigma\ [1/(n+1) - 1/(n+2)] = 4 \cdot (1/2 - 0) = 2$

Answer: E[X] = 2 (QED)

(c) Show that $E[X^2] = \infty$

$E[X^2] = \Sigma\ n^2 \cdot 4/[n(n+1)(n+2)] = 4\ \Sigma\ n/[(n+1)(n+2)]$

For $n \geq 1$ we have $(n+1)(n+2) \leq (2n)(3n) = 6n^2$, so

$n/[(n+1)(n+2)] \geq n/(6n^2) = 1/(6n)$

Therefore $E[X^2] \geq (4/6)\ \Sigma\ 1/n = (2/3)\ \Sigma\ 1/n$, and the harmonic series diverges. By comparison:

Answer: $E[X^2] = \infty$ (QED) (so Var(X) is also infinite)

Problem 3 (4 points)

$f(x) = a + bx^2$ on [0, 1]. Two conditions determine a and b.

Condition 1 (total probability is 1):

$\int_0^1 (a + bx^2)\ dx = a + b/3 = 1$

Condition 2 (E[X] = 3/5):

$\int_0^1 x(a + bx^2)\ dx = a/2 + b/4 = 3/5$

From Condition 1, $a = 1 - b/3$. Substituting:

$(1 - b/3)/2 + b/4 = 3/5 \implies 1/2 - b/6 + b/4 = 3/5 \implies b/12 = 1/10 \implies b = 6/5$

$a = 1 - (6/5)/3 = 1 - 2/5 = 3/5$

Answer: a = 3/5 = 0.6000 and b = 6/5 = 1.2000

Problem 4 (4 points)

Let $Y = (X - \mu)/\sigma = (1/\sigma)X - \mu/\sigma$, with $E[X] = \mu$ and $Var(X) = \sigma^2$.

$E[Y] = (E[X] - \mu)/\sigma = (\mu - \mu)/\sigma = 0$

$Var(Y) = Var(X/\sigma - \mu/\sigma) = (1/\sigma^2) \cdot Var(X) = \sigma^2/\sigma^2 = 1$

(Subtracting the constant μ/σ does not change the variance, and multiplying by $1/\sigma$ multiplies the variance by $1/\sigma^2$.)

Answer: E[Y] = 0 and Var(Y) = 1

Problem 5 (8 points)

$X \sim \text{Poisson}(\lambda_1 = 2)$, $Y \sim \text{Poisson}(\lambda_2 = 3)$, independent, $W = X + Y$.

(a) MGF of W

The MGF of a Poisson(λ) variable is $M(t) = \Sigma\, e^{tk}\, e^{-\lambda}\, \lambda^k/k! = \exp(\lambda(e^t - 1))$. So

$M_X(t) = \exp(2(e^t - 1))$, $M_Y(t) = \exp(3(e^t - 1))$

For independent X and Y, the MGF of the sum is the product of the MGFs:

$M_W(t) = M_X(t) \cdot M_Y(t) = \exp(2(e^t - 1) + 3(e^t - 1)) = \exp(5(e^t - 1))$

Answer: $M_W(t) = \exp(5(e^t - 1))$

(b) Distribution of W

This is exactly the MGF of a Poisson random variable with parameter 5. By the uniqueness property of MGFs, W has that distribution.

Answer: W ~ Poisson(λ = 5), i.e. $P(W = k) = e^{-5} \cdot 5^k/k!$, k = 0, 1, 2, …

(c) E[W] and $E[W^2]$

Differentiate $M_W(t) = \exp(5(e^t - 1))$:

$M'_W(t) = 5e^t \cdot \exp(5(e^t - 1)) = 5e^t \cdot M_W(t)$

$M''_W(t) = 5e^t \cdot M_W(t) + 5e^t \cdot M'_W(t) = 5e^t M_W(t) + 25e^{2t} M_W(t)$

Evaluate at t = 0, where $M_W(0) = e^0 = 1$:

$E[W] = M'_W(0) = 5 \cdot 1 \cdot 1 = 5$

$E[W^2] = M''_W(0) = 5 + 25 = 30$

Answer: E[W] = 5.0000 and $E[W^2]$ = 30.0000 (so Var(W) = 30 − 25 = 5, matching Poisson(5))